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**LEADING THE WORLD**



AN INTERNATIONAL AWARD-WINNING INSTITUTION FOR SUSTAINABILITY

## KAGGLE COMPETITION REPORT

### MCTA 4363:Deep Learning

### LLMs: You Can't Please Them All

LECTURERS: Dr. Hasan Firdaus Bin Mohd Zaki

| NAME                        | MATRIC NO. |
|-----------------------------|------------|
| AHMAD HAFIZI BIN ISHAK      | 2111775    |
| LUQMAN AZFAR BIN AZMI       | 2219857    |
| MUHAMMAD AFIQ BIN MOHD ASRI | 2212541    |

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## Introduction

The rapid advancement of Large Language Models (LLMs) has enabled machines to generate highly coherent and human-like text. However, evaluating such generated content remains challenging due to subjective human preferences. The Kaggle competition “*LLMs: You Can’t Please Them All*” focuses on modeling disagreements among evaluators when assessing AI-generated responses.

This project implements an adversarial evaluation framework where AI generates essays designed to maximize disagreement among multiple judges.

## 2.Problem Statement

Different evaluators often have conflicting opinions when assessing text quality. This makes it difficult to define a single "correct" answer.

The problem addressed in this project is:

How can disagreement among multiple evaluators be modeled and quantified for AI-generated text?

## 3.Objectives

- Generate adversarial essays using LLM strategies
- Simulate multi-judge evaluation
- Measure disagreement using statistical metrics
- Compute a final system score

## 4. Methodology

### 4.1 Overall System Architecture

The proposed system follows a two-stage pipeline consisting of a **generation module** and an **evaluation module**. The goal is to intentionally generate essays that expose disagreement among evaluators and then quantify that disagreement using statistical measures.

Flow of the system:

1. Input topic and adversarial strategy
2. Essay generation using LLM (Gemini 3 Flash)
3. Multi-judge evaluation
4. Statistical analysis (variance, gap, averages)
5. Final system score computation

### 4.2 Generation Phase (Adversarial Essay Creation)

The generation phase acts as the *player* in the system. It uses prompt engineering to manipulate the writing style of the LLM.

Three adversarial strategies are defined:

- The Contradictor  
Forces the model to present conflicting ideas simultaneously, reducing logical coherence.
- The Jargon-Bomb  
Uses dense academic vocabulary to increase perceived sophistication while reducing readability.

- The Emotional-Hook

Focuses on strong metaphors and emotional intensity to create subjective interpretation differences.

The prompt is dynamically constructed based on the selected strategy:

Prompt = f(Topic, Strategy)

This ensures variation in outputs and simulates real-world prompt engineering techniques.

4.3 Evaluation Phase (Multi-Judge Simulation)

The evaluation phase represents a **committee of judges**, each with different biases and evaluation styles.

Judges Configuration

| Judge | Characteristics         |
|-------|-------------------------|
| J1    | Strict, grammar focused |
| J2    | Emotion, sensitive      |
| J3    | Logical and structured  |

Each judge assigns scores from 0 to 10 across five criteria:

- Grammar & Vocabulary
- Coherence
- Development
- Content
- Emotion

## 4.4 Bias Simulation Mechanism

To reflect realistic disagreement, the system introduces **controlled bias**:

- Emotional strategy heavily affects the *Emotion* criterion
- J2 tends to give higher emotion scores
- J1 penalizes inconsistency
- J3 prefers logical clarity

Example:

- Emotion scores: [2.0, 9.8, 4.5]

This creates large variation, simulating real evaluator disagreement.

## 4.5 Statistical Analysis

For each evaluation criterion, the following metrics are computed:

### (a) Mean Score (Quality)

$$Q = (1/n) * \sum(x_i)$$

Represents overall performance across judges.

### (b) Variance (Disagreement)

$$V = \sqrt{(1/n) * \sum((x_i - Q)^2)}$$

Measures how much judges disagree.

### **(c) Pairwise Differences**

- $J1 - J2$
- $J2 - J3$
- $J1 - J3$

Used to identify which judges disagree the most.

### **(d) Maximum Gap**

$$\text{Max Gap} = \max(x_i) - \min(x_i)$$

Highlights extreme disagreement in scoring.

## **4.6 Final Score Computation**

The final system score combines both quality and disagreement:

$$\text{Final Score} = V * (9 - Q)$$

Where:

- $Q$  = average quality score
- $V$  = average variance across all criteria

Interpretation:

- High quality + low disagreement  $\rightarrow$  lower score
- High disagreement  $\rightarrow$  higher score

This aligns with the competition objective of identifying outputs that cannot please everyone.

## 4.7 Implementation Tools

- Python for system development
- NumPy for statistical computation
- Random module for score simulation
- Pandas for data handling
- Google Gemini API for essay generation

## 4.8 System Overview

The system consists of two phases:

### (A) Generation Phase

- Uses Gemini 3 Flash model

### (B) Evaluation Phase

Three simulated judges:

- J1 (Strict)
- J2 (Emotional)
- J3 (Logical)

Evaluation Criteria:

- Grammar & Vocabulary
- Coherence
- Development
- Content



- Emotion

## 5. Results

### 5.1 Table Comparison

| Strategy           | Conflict Criterion | Max Gap | Quality Score | Variance | Final System Score | Observation                                                            |
|--------------------|--------------------|---------|---------------|----------|--------------------|------------------------------------------------------------------------|
| The Contradictor   | Coherence          | 7.7     | 6.21          | 1.24     | 3.4596             | Generated the strongest disagreement among judges                      |
| The Jargon-Bomb    | Grammar & Vocab    | 6.5     | 6.40          | 0.80     | 2.0800             | Had the most stable evaluation overall, though judges                  |
| The Emotional-Hook | Emotion            | 7.8     | 6.34          | 1.06     | 2.8196             | Triggered extreme emotional interpretation differences between judges. |

## 5.2 Results

| <div> KAGGLE ADVERSARIAL TRIAL: The Emotional-Hook</div> <div>TOPIC: Self-reliance and Adaptability in Healthcare</div> <div>=====</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------|------------|------------|---------|----------|-------------------|-------------------------|--|--|---------|-----------------|-----------------|------------|------------|------------|-----|-----------|-----------------|------------|------------|------------|-----|-------------|-----------------|------------|------------|------------|-----|---------|-----------------|------------|------------|------------|-----|---------|-----------------|------------|------------|------------|-----|
| <div> AI-GENERATED ESSAY:</div> <div>In white-tiled trenches, the healer stands as a solitary bulwark against encroaching void. Self-reliance is the jagged flint struck in darkness, sparking survival when all systems crumble. We are chameleons in a furnace, shedding skin to breathe amidst the choking smoke. When a pulse falters, adaptability is the desperate dance of a willow in a gale, bending, never breaking. In this sanctuary of screams, we are forged steel, eternally evolving or meeting a cold, silent end. This is our war.</div>                                                                                                                                                                                                                                                                |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
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| CRITERIA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SCORES (J1,J2,J3) | INTER-JUDGE DIFFERENCES |            |            | MAX GAP |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Grammar & Vocab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [5.9, 6.4, 6.5]   | J1-J2: 0.5              | J2-J3: 0.1 | J1-J3: 0.6 | 0.6     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Coherence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [6.6, 6.3, 5.6]   | J1-J2: 0.3              | J2-J3: 0.7 | J1-J3: 1.0 | 1.0     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [6.5, 7.8, 7.6]   | J1-J2: 1.3              | J2-J3: 0.2 | J1-J3: 1.1 | 1.3     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [7.6, 6.4, 5.6]   | J1-J2: 1.2              | J2-J3: 0.8 | J1-J3: 2.0 | 2.0     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Emotion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [2.0, 9.8, 4.5]   | J1-J2: 7.8              | J2-J3: 5.3 | J1-J3: 2.5 | 7.8     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> FINAL SYSTEM SCORE: 2.8196 (Variance: 1.06, Quality: 6.34)</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div>=====</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> KAGGLE ADVERSARIAL TRIAL: The Jargon-Bomb</div> <div>TOPIC: Self-reliance and Adaptability in Healthcare</div> <div>=====</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> AI-GENERATED ESSAY:</div> <div>The praxis of iatritic paradigms necessitates an autarkic agency, where endogenous resilience facilitates heuristic autonomy amidst volatile nosological shifts. Clinicians must embrace polymorphic synergetics, transcending ossified protocols via homeostatic recalibration. This teleological flux demands epistemic versatility, ensuring the intra-subjective praxis of care remains robust against systemic entropy. By synthesizing neuroplastic malleability with deontological individualization, practitioners navigate the clinico-pathological landscape, achieving ontological equanimity that mitigates profoundly the stochastic perturbations inherent in contemporary biomedical exigencies and the multi-faceted, phenomenological therapeutic landscapes.</div> |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
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| CRITERIA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SCORES (J1,J2,J3) | INTER-JUDGE DIFFERENCES |            |            | MAX GAP |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Grammar & Vocab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [9.5, 6.0, 3.0]   | J1-J2: 3.5              | J2-J3: 3.0 | J1-J3: 6.5 | 6.5     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Coherence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [6.4, 6.3, 6.1]   | J1-J2: 0.1              | J2-J3: 0.2 | J1-J3: 0.3 | 0.3     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [6.5, 5.2, 7.0]   | J1-J2: 1.3              | J2-J3: 1.8 | J1-J3: 0.5 | 1.8     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [5.9, 6.0, 6.6]   | J1-J2: 0.1              | J2-J3: 0.6 | J1-J3: 0.7 | 0.7     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Emotion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [7.0, 7.4, 7.1]   | J1-J2: 0.4              | J2-J3: 0.3 | J1-J3: 0.1 | 0.4     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> FINAL SYSTEM SCORE: 2.08 (Variance: 0.8, Quality: 6.4)</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div>=====</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> KAGGLE ADVERSARIAL TRIAL: The Contradictor</div> <div>TOPIC: Self-reliance and Adaptability in Healthcare</div> <div>=====</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> AI-GENERATED ESSAY:</div> <div>To achieve self-reliance in healthcare, one must strictly depend on everyone else. True adaptability means refusing to change so that everything stays different. A provider who stands alone is actually surrounded by a crowd of isolation. By following rigid protocols flexibly, we ensure that progress remains completely still. To heal, we must ignore the patient to focus entirely on their needs. Only through static motion can we find the stable chaos required for permanent temporary wellness. This is modern medicine.</div>                                                                                                                                                                                                                                     |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
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| CRITERIA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SCORES (J1,J2,J3) | INTER-JUDGE DIFFERENCES |            |            | MAX GAP |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Grammar & Vocab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [5.9, 6.6, 7.5]   | J1-J2: 0.7              | J2-J3: 0.9 | J1-J3: 1.6 | 1.6     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Coherence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [4.5, 9.2, 1.5]   | J1-J2: 4.7              | J2-J3: 7.7 | J1-J3: 3.0 | 7.7     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [5.3, 7.4, 7.6]   | J1-J2: 2.1              | J2-J3: 0.2 | J1-J3: 2.3 | 2.3     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [6.2, 5.0, 6.6]   | J1-J2: 1.2              | J2-J3: 1.6 | J1-J3: 0.4 | 1.6     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| Emotion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [6.5, 7.5, 5.9]   | J1-J2: 1.0              | J2-J3: 1.6 | J1-J3: 0.6 | 1.6     |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div> FINAL SYSTEM SCORE: 3.4596 (Variance: 1.24, Quality: 6.21)</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |
| <div>=====</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                         |            |            |         |          |                   |                         |  |  |         |                 |                 |            |            |            |     |           |                 |            |            |            |     |             |                 |            |            |            |     |         |                 |            |            |            |     |         |                 |            |            |            |     |

## **6.Discussion**

The experimental results show that each adversarial strategy produces different patterns of evaluator disagreement. This supports the main objective of the competition, which is to identify responses that cannot satisfy all evaluators equally.

Among the three strategies, The Contradictor achieved the highest final system score (3.4596). The use of conflicting statements created major disagreement in the Coherence criterion, showing that logical inconsistency strongly affects evaluator perception.

The Jargon-Bomb strategy obtained the highest average quality score ( $Q = 6.40$ ) but the lowest variance. Although the complex academic vocabulary increased perceived sophistication, evaluator opinions remained relatively consistent compared to the other strategies.

The Emotional-Hook strategy generated the largest disagreement in the Emotion criterion, with a maximum gap of 7.8. Emotional and metaphorical language caused highly subjective interpretations between judges.

Overall, subjective criteria such as Emotion and Coherence produced larger disagreements than objective criteria like Grammar & Vocabulary. The findings also show that higher writing quality does not necessarily lead to higher adversarial performance. Instead, strategies that maximize evaluator inconsistency achieve better final scores.

This experiment demonstrates that adversarial prompt engineering can effectively manipulate evaluator perception and create measurable disagreement patterns. The results strongly support the central idea of the Kaggle competition: no single response can please all evaluators simultaneously.

## **7.Future works**

- Replace simulated judges with real Large Language Models (LLMs)
- Experiment with additional adversarial prompting strategies such as sarcasm, ambiguity, misinformation, humor, and mixed writing styles to generate stronger evaluator inconsistency.
- extend system into a multi-model framework such as Gemini, GPT, and Claude for essay generation.
- measure judge disagreement and model robustness across different LLM outputs.